Experiment 6: End-to-End Data Pipeline Orchestration using Azure Data Factory and Azure Databricks

Course: Cloud Analytics (25MCAR0141) | Course Outcome: CO3

Aim

To integrate Azure Data Factory with Azure Databricks for end-to-end data pipeline orchestration, where ADF triggers a Databricks notebook to preprocess and transform data ingested from Azure Data Lake Storage Gen2.

Objectives

- Create ADF Linked Services for Blob Storage, ADLS Gen2, and Azure Databricks.
- Build an ADF pipeline with Copy Data activity followed by Databricks Notebook activity.
- Configure success-dependency chaining between pipeline activities.
- Implement PySpark data cleaning and feature engineering in a Databricks notebook.
- Validate the transformed Parquet output in ADLS Gen2.
- Monitor pipeline runs and activity status in ADF Studio.

Architecture & Component Roles

Resource Group: 25MCAR0141

Logical container for all Azure experiment resources

Data Factory: calab-adf-141

Cloud data orchestration engine executing the ELT pipeline

Storage Account: calablake141

ADLS Gen2 hosting rawdata (input) and output (transformed) containers

Databricks Workspace: calab-databricks-141

Apache Spark platform executing the TransformData notebook

Linked Service: LS_Databricks

ADF to Databricks connection (MSI auth, job cluster)

Pipeline: PL_ADF_Databricks

Orchestrates Copy Data -> Databricks Notebook activities

Architecture Diagram

```
Azure Blob Storage (student_performance.csv)
        |  Copy Data activity
        v
ADLS Gen2 (calablake141 / rawdata)
        |  Databricks Notebook activity (on Copy success)
        v
Azure Databricks (calab-databricks-141)
        |  TransformData notebook: clean + feature engineering
        v
ADLS Gen2 (calablake141 / output)
        v
output/student_performance_cleaned (Parquet)
```

Step-by-Step Execution Summary

- Step 1: Open ADF Studio: Azure Portal -> calab-adf-141 -> Launch Studio.
- Step 2: Create Linked Service - Blob Storage: Manage -> Linked Services -> + New -> Azure Blob Storage -> name LS_BlobSource -> select calablake141.
- Step 3: Create Linked Service - ADLS Gen2: + New -> Azure Data Lake Storage Gen2 -> name LS_ADLS_Sink -> select calablake141.
- Step 4: Create Linked Service - Databricks: + New -> Azure Databricks -> name LS_Databricks -> workspace URL: adb-7405609713574305.5.azuredatabricks.net -> New job cluster -> Runtime 14.3, Standard_DS3_v2, 1 worker.
- Step 5: Create Source Dataset: Author -> Datasets -> + New -> DelimitedText -> name DS_BlobSource_CSV -> linked service LS_BlobSource -> file student_performance.csv.
- Step 6: Create Sink Dataset: + New -> DelimitedText -> name DS_ADLS_Sink_CSV -> linked service LS_ADLS_Sink -> container rawdata.
- Step 7: Upload TransformData Notebook: In Databricks Studio, import TransformData.py as a notebook.
- Step 8: Create Pipeline: Author -> Pipelines -> + New -> name PL_ADF_Databricks.
- Step 9: Add Copy Data Activity: Drag Copy Data -> Source: DS_BlobSource_CSV -> Sink: DS_ADLS_Sink_CSV.
- Step 10: Add Databricks Notebook Activity: Drag Databricks Notebook -> connect after Copy Data (success dependency) -> select LS_Databricks -> browse notebook /Users/.../TransformData.
- Step 11: Validate & Publish: Click Validate -> Publish All.
- Step 12: Trigger Pipeline: Add Trigger -> Trigger Now -> run the pipeline.
- Step 13: Monitor & Verify: Monitor -> Pipeline Runs -> verify both activities show Succeeded. Check calablake141/output/student_performance_cleaned for Parquet files.

Pipeline Activity Chain

```
CopyRawDataToADLS (Copy Data)
    |
    |-- Source: DS_BlobSource_CSV (Blob Storage)
    |-- Sink:   DS_ADLS_Sink_CSV (ADLS Gen2 rawdata)
    |
    +-- On Success --> RunTransformDataNotebook (Databricks Notebook)
                           |
                           |-- Linked Service: LS_Databricks
                           |-- Notebook: /Users/.../TransformData
                           +-- Output: ADLS Gen2 output/ (Parquet + CSV)
```

TransformData Notebook - Highlights

Cleaning: strip whitespace from column names and string values, drop duplicate rows, fill numeric nulls with the column mean and string nulls with 'Unknown'.

Feature engineering (score columns detected by name): total_score, average_score, performance_band (Excellent >= 85, Good >= 70, Average >= 50, otherwise Needs Improvement), result (Pass if average >= 40, otherwise Fail).

Output: Parquet to output/student_performance_cleaned, plus a single CSV copy to output/student_performance_cleaned_csv.

Pipeline and linked-service JSON definitions are in pipeline/PL_ADF_Databricks.json and linkedService/LS_Databricks.json in the repository folder.

Screenshot Checklist

- 1. ADF Studio - Linked Services list (LS_BlobSource, LS_ADLS_Sink, LS_Databricks) (Step_01_*.png - see README for exact filenames)
- 2. LS_Databricks configuration (workspace URL, cluster config) (Step_02_*.png - see README for exact filenames)
- 3. Dataset DS_BlobSource_CSV configuration (Step_03_*.png - see README for exact filenames)
- 4. Pipeline canvas - Copy Data -> Databricks Notebook chain (Step_04_*.png - see README for exact filenames)
- 5. Copy Data activity settings (source & sink) (Step_05_*.png - see README for exact filenames)
- 6. Databricks Notebook activity settings (notebook path) (Step_06_*.png - see README for exact filenames)
- 7. Pipeline validation - no errors (Step_07_*.png - see README for exact filenames)
- 8. Trigger Now - pipeline triggered (Step_08_*.png - see README for exact filenames)
- 9. Monitor - Pipeline run succeeded (both activities green) (Step_09_*.png - see README for exact filenames)
- 10. Copy Data activity run details (Step_10_*.png - see README for exact filenames)
- 11. Databricks Notebook activity run details (with output) (Step_11_*.png - see README for exact filenames)
- 12. ADLS Gen2 - rawdata container with uploaded CSV (Step_12_*.png - see README for exact filenames)
- 13. ADLS Gen2 - output/student_performance_cleaned Parquet files (Step_13_*.png - see README for exact filenames)
- 14. Databricks notebook - TransformData output display (Step_14_*.png - see README for exact filenames)

Result

The ADF pipeline ran end to end: Copy Data moved student_performance.csv from Blob Storage into ADLS Gen2 rawdata, and on its success the Databricks Notebook activity ran TransformData, cleaning the data and adding score features, with the transformed output validated as Parquet files in calablake141/output/student_performance_cleaned. Both activities showed Succeeded in ADF Monitor.

Key Learning

This experiment demonstrates ELT orchestration on Azure: ADF as the control plane (linked services, datasets, success-dependency chaining, monitoring) and Databricks as the compute plane (PySpark cleaning and feature engineering), with ADLS Gen2 as the shared raw and transformed storage layer between them.